

# [70]

## **Project Title: Cucumber Slicer Mechanism.**

### **Objective**

The project aims to design and implement an automated slicing cucumbers system.

### **Project Description**

- The system is built on a stable, ground-mounted frame that supports the cutting unit, feeding mechanism, and safety features.
- Cucumbers are loaded into a magazine at the start of the process.
- A sensor detects the presence of a cucumber at the bottom of the magazine, triggering the slicing sequence.
- The mechanical system consists of three pneumatic cylinders operating in a sequential manner:
  - 1- Cylinder 1: Activated in a horizontal motion to push the cucumber from the magazine to the slicing position.
  - 2- Cylinder 2: Activated in a vertical motion to perform the slicing. This cylinder is equipped with a plate mounted with multiple blades to cut the cucumber into several pieces in one stroke.
  - 3- Cylinder 3: Activated in a horizontal motion to push the sliced cucumber onto a collection tray.
- The operation is controlled using a relay logic based on sensor feedback and predefined logic.
- The system is designed for continuous operation, with repeatable and safe slicing cycles.

The description above outlines the core operation of the system. Identify and include all necessary components required for proper system operation, such as actuators, sensors, and control elements, to ensure the process operates continuously through multiple cycles.